

Introduction to Statistics for Data Science

Descriptive statistics summarize a dataset using measures like mean, median, mode, variance, and standard deviation.

Probability distributions describe how likely different outcomes are. The normal distribution is symmetric and bell shaped.

Hypothesis testing evaluates whether observed data supports a claim about a population. A null hypothesis represents the status quo.

Confidence intervals provide a range of plausible values for a population parameter, calculated from sample data. A 95% confidence interval means that if we repeated the study many times, 95% of the intervals would contain the true parameter value.

Correlation measures the strength and direction of a linear relationship between two variables, ranging from negative one to positive one.

Regression analysis models the relationship between a dependent variable and one or more independent variables. Linear regression is the most common type.

The central limit theorem states that the sampling distribution of the sample mean approaches a normal distribution as the sample size increases.

Key vocabulary: mean, median, standard deviation, normal distribution, p value, confidence interval, correlation, regression.